EQUALITY OF DIMENSIONS AND REDUCEDNESS OF THE RESIDUE FIELD TENSOR PRODUCT FORCE THE MAXIMAL IDEALS TO AGREE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof contains the author's parenthetical justification of the separability cases; it is reproduced verbatim and the cited result is carried as a navigation stub.

Editorial note. The counted spans carry 4 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-change-fields-algebraic-unramified is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Structure of the local rings under a ground field extension. Same chapter, Lemma lemma-change-fields-flat, invoked at proof lines 312-313 and named in the statement. Supplies the description of the larger local ring as a localization of the tensor product and the identification of its residue field.
- Dimension computation for the ground field change. Same chapter, Lemma lemma-change-fields-algebraic-dim, invoked at proof lines 312-313. Supplies that the quotient by the extended maximal ideal has dimension zero.
- A separable extension preserves reducedness, so the parenthetical cases are covered. Algebra, Lemma algebra-lemma-separable-extension-preserves-reducedness, invoked at proof lines 310-311. Supplies the parenthetical statement that separability implies the reducedness hypothesis.

Source reference. Original lemma 030U: sources/stacks-chow/tags.txt, source lines 3961–3961. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0C4Z: sources/stacks-varieties/pinned-original-source.tex, source lines 284–284. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0C4Y: sources/stacks-varieties/pinned-original-source.tex, source lines 238–238. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Notation as in Lemma 0C4Y. Assume X is locally of finite type over k, that $\dim(\mathcal{O}_{X,x}) = \dim(\mathcal{O}_{Y,y})$ and that $\kappa(x) \otimes_k K$ is reduced (for example if $\kappa(x)/k$ is separable or K/k is separable). Then $\mathfrak{m}_x\mathcal{O}_{Y,y} = \mathfrak{m}_y$.*

Proof. (The parenthetical statement follows from Algebra, Lemma 030U.) Combining Lemmas 0C4Y and 0C4Z we see that $\mathcal{O}_{Y,y}/\mathfrak{m}_x\mathcal{O}_{Y,y}$ has dimension 0 and is reduced. Hence it is a field. □